

Dr. Babasaheb Ambedkar Technological University, Lonere

Regular and Supplementary Winter Examination – 2023

Course: B. Tech.

Branch : Civil Engineering

Semester :V

Subject Code & Name: BTCVC502 Geotechnical Engineering

Max Marks: 60

Date: 03/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Mark
		s
Q. 1 Solve <u>any two</u> of the following.		12
A) Explain the Following terms:	Understand/ CO1	6
A.1) Voids Ratio A.2) Porosity A.3) Degree of saturation		
A.4) Water Content A.5) Specific Gravity A.6) Air Content		
B) Why soil is considered as three phase system? Explain with neat sketch?	Understand/ CO1	6
C) State and explain the different types of soil structures?	Understand/ CO1	6
Q.2 Solve <u>any two</u> of the following.		12
A) Enlist the different soil classification systems that were developed by various organizations for their specific purpose. Explain in detail the by 'Indian Soil Classification system'.	CO2/ Apply	6
B) A soil sample of saturated clay has a diameter of 50 mm and the height of 100 mm. The mass of saturated sample is 220 g and its mass when oven dried is 150 g. Find: 1) Water content of the clay & 2) Void Ratio	Understand / CO1	6
C) A soil sample has wet density 20 kN/m ³ and dry density of 18 kN/m ³ . If the specific gravity of soil is 2.67. Calculate the porosity, moisture content and degree of saturation. Assume unit weight of water = 10 kN/m ³	Apply /CO2	6
Q. 3 Solve <u>any two</u> of the following.		12
A) What is mean by permeability of soil. Explain in detail the concept of Hydraulic gradient and Darcy Law?	Understand / CO2	6
B) Determine the average coefficient of permeability in the horizontal and vertical directions for a deposit consisting of three layers of thickness 5 m, 1 m and 2.5 m and having the coefficients of permeability of 3×10^{-2} mm/sec, 3×10^{-5} mm/sec, and 4×10^{-2} mm/sec, respectively. Assume the layers are isotropic.	Apply /CO3	6
C) In falling head permeability test, the head causing flow was initially 90 cm, and it drops 6 cm in 15 minutes. How much time is required for the head to fall to 45 cm.	Apply /CO3	6

Q.4 Solve any two of the following. **12**

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| A) Explain the fundamental equation of shear strength of soil given by Coulomb. | Understand
/ CO2 | 6 |
| B) Explain the different types of shear test based on drainage conditions. | Apply
/ CO3 | 6 |
| C) A vane 100 mm in diameter and 200 mm in heights was pressed into soft clay in a borehole. The torque was applied and gradually increased to 100 N-m when failure took place. Determine the undrain shear strength. | Understand
/ CO2 | 6 |

Q. 5 Solve any two of the following. **12**

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| A) Explain in detail the major factors which affect the compaction? | Understand/
CO2 | 6 |
| B) An earth embankment is compacted at a water content of 12 % to a bulk density of 25 kN/m ³ . If the specific gravity of the sand is 2.65, find the porosity, void ratio and the degree of saturation of the compacted embankment. | Understand
/ CO2 | 6 |
| C) A gravity retaining wall retains 10 m of a backfill, $\gamma = 25 \text{ kN/m}^3$ $\phi = 20^\circ$ with a uniform horizontal surface. Determine the total active thrust per meter run of wall. Assume the wall interface to be vertical. | Apply
/ CO3 | 6 |

***** End *****